

David Hoyos Alzate

AI Product Builder · Mechanical Engineer

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PROFILE

Mechanical engineer turned AI product builder. I turn manual processes into agents, automations and internal tools that reach production —directing AI with product judgment, tests and clean handoff documentation—. I bring an engineer's rigor (I measure what I build) and a builder's speed with AI (I ship in days, not months). Trained at Builders, the AI-building program by Andrés Bilbao.

LINKS

GitHub: github.com/dahoyosa6 · LinkedIn: linkedin.com/in/davidhoyosalzate · Portfolio: david-hoyos.vercel.app

EXPERIENCE

2026 – Present

De cero a uno — AI Product Builder

Own studio · Remote

- Built Saku, a WhatsApp sales agent for SMBs that handles price, availability and catalog questions 24/7; it answered correctly in ~80% of a custom suite of 84 test conversations evaluated automatically. I designed it multi-tenant from day one, with the engine separated from the configuration, so adapting it to another company is a matter of changing config, not rewriting code (pre-pilot with a real client in Mexico). Stack: n8n · Supabase + pgvector · WhatsApp Cloud API · Python.
- Centralized 16 accounts with immediate visibility into their status by shipping FinanzasIQ to production, a personal-finance system: a pipeline (bank emails + OCR → n8n → Google Sheets → a ~13,000-line dashboard) captures most transactions automatically and cuts logging a transaction down to seconds. I implemented real operational security (OTP access, JWT validation and server-side secrets).
- Shipped 3 more products live: Momentum (planner, PWA, in production), Nexo (matching of 334 scholarships with embeddings) and AskDB (chat-based data analyst: natural language → safe SQL query + voice answer). Hundreds of automated tests as a safety net.

Jul 2025 – Present

Mitsubishi Electric de Colombia — Product Engineer

Medellín

- Halved the time to assign an order by actively taking part in automating the elevator-component assignment process, which became standardized and required less manual work.
- Sped up structural calculation and made it consistent across the team by developing templates that standardize the calculation process and quickly show whether a component passes or not.
- Handle special customer requests with custom designs —for example, a cabin with an infinity-tunnel visual effect (optics and LED lighting).

Jan – Jul 2025

Mitsubishi Electric de Colombia — R&D&i Intern

Medellín

- Enabled staff safety training by designing, building and installing the Safety Experience Center (CES) with functional mechanical modules.
- Backed design decisions with data by validating component strength through finite element analysis (FEA).
- Optimized plant tooling by designing custom parts with CAD software.

Jul – Dec 2024

Tribos Ingeniería S.A.S. — Assistant, SIPRALUB Laboratory

Medellín

- Supported machinery condition diagnosis by detecting incipient failures, measuring mechanical vibrations and analyzing frequency spectra; programmed and operated accelerometers for data capture.
- Performed mechanical maintenance tasks (alignment, balancing and lubrication of components).

SKILLS

Applied AI & agents: Claude / LLMs, prompt engineering, RAG (pgvector), agent evaluation, safe text-to-SQL

Automation & integrations: n8n, REST APIs, WhatsApp Cloud API, Telegram, webhooks, OCR

Development (AI-first): I build and ship software by directing AI —Python (FastAPI), TypeScript / JavaScript, SQL— with my own architecture, tests and refactoring

Data & cloud: PostgreSQL (Supabase, Neon), Google Sheets / Apps Script, Vercel, Railway, Cloudflare, Git / GitHub (Actions)

Product & systems: MVP definition, architecture design, handoff documentation, operational security (JWT, RLS, secrets management)

Mechanical engineering: CAD/CAE (SolidWorks, Solid Edge, Fusion, Ansys), finite element analysis (FEA)

EDUCATION

Universidad Nacional de Colombia — Mechanical Engineering · 2019 – 2025

Builders — Building products with AI (Andrés Bilbao) · 2026

LANGUAGES

Spanish (native) · English (advanced)